A First Course in Causal Inference Ch. 11

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Ch. 12 The Doubly Robust or the Augmented Inverse Propensity Score Weighting Estimator for the Average Causal Effect

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Assumption 12.0

Ignorability $Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid X$ and overlap $0 < e(X) < 1$.

Under Assum. 12.0, we have two identification formula of the average causal effect $\tau = \mathbb{E}[Y(1) - Y(0)]$. First, the outcome regression formula is

$$\tau = \mathbb{E}[\mu_1(X)] - \mathbb{E}[\mu_0(X)], \tag{1}$$

where,

$$\mu_1(X) = \mathbb{E}[Y(1) \mid X] = \mathbb{E}[Y \mid Z = 1, X]$$
$$\mu_0(X) = \mathbb{E}[Y(0) \mid X] = \mathbb{E}[Y \mid Z = 0, X].$$

Second, the IPW formula is

$$\tau = \mathbb{E}\left[\frac{ZY}{e(X)}\right] - \mathbb{E}\left[\frac{(1-Z)Y}{1-e(X)}\right], \tag{2}$$

where $e(X) = \mathbb{E}[Z = 1 \mid X]$. Eq. (1) is consistent if the outcome model is correctly specified, and Eq. (2) is consistent if the propensity score model is correctly specified.

- What if one of them is not correctly specified?
- We are then motivated to propose an estimator that is consistent if either of them is correctly specified.
- It is called the *doubly robust* estimator (Scharfstein et al., 1999; Bang and Robins, 2005).

12.1 The doubly robust estimator

- We posit a model for the conditional means of the outcome $\mu_1(X, \beta_1), \mu_0(X, \beta_0)$, with parameters β_1, β_0.
- If the outcome model is corectly specified, then $\mu_1(X) = \mu_1(X, \beta_1)$ and $\mu_0(X) = \mu_0(X, \beta_0)$.
- We posit a model for the propensity score $e(X, \alpha)$, with a parameter α.
- If the propensity score model is correctly specified, then $e(X) = e(X, \alpha)$.
- In practice, both models may be misspecified.
- Sometimes, we call them *working models* due to the possibility of misspecification.

12.1 The doubly robust estimator

Define

$$\tilde{\mu}_1^{\mathsf{dr}} = \mathbb{E}\left[\frac{Z\{Y-\mu_1(X,\beta_1)\}}{e(X,\alpha)} + \mu_1(X,\beta_1)\right], \tag{3}$$

$$\tilde{\mu}_0^{\mathsf{dr}} = \mathbb{E}\left[\frac{(1-Z)\{Y-\mu_0(X,\beta_0)\}}{1-e(X,\alpha)} + \mu_0(X,\beta_0)\right], \tag{4}$$

which can be written as

$$\tilde{\mu}_1^{\mathsf{dr}} = \mathbb{E}\left[\frac{ZY}{e(X,\alpha)} - \frac{Z-e(X,\alpha)}{e(X,\alpha)}\mu_1(X,\beta_1)\right], \tag{5}$$

$$\tilde{\mu}_0^{\mathsf{dr}} = \mathbb{E}\left[\frac{(1-Z)Y}{1-e(X,\alpha)} - \frac{e(X,\alpha)-Z}{1-e(X,\alpha)}\mu(X,\beta_0)\right]. \tag{6}$$

12.1 The doubly robust estimator

- Eq. (3), (4) augment the outcome regression estimator by inverse propensity score weighting terms of the residuals $(Y - \mu_Z(X, \beta_Z))$.
- Eq. (5), (6) augment the IPW estimator by the imputed outcomes.
- For this reason, the doubly robust estimator is also called the *augmented inverse propensity score weighting* (AIPW) estimator.
- The augmentation strengthens the theoretical properties in the following sense.

Theorem 12.1

Assume ignorability $Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid X$ and overlap $0 < e(X) < 1$.

1. If either $e(X, \alpha) = e(X)$ or $\mu_1(X, \beta_1) = \mu_1(X)$, then $\tilde{\mu}_1^{\mathsf{dr}} = \mathbb{E}[Y(1)]$.
2. If either $e(X, \alpha) = e(X)$ or $\mu_0(X, \beta_1) = \mu_0(X)$, then $\tilde{\mu}_0^{\mathsf{dr}} = \mathbb{E}[Y(0)]$.
3. If either $e(X, \alpha) = e(X)$ or $\{\mu_1(X, \beta_1) = \mu_1(X), \mu_0(X, \beta_1) = \mu_0(X)\}$, then $\tilde{\mu}_1^{\mathsf{dr}} - \tilde{\mu}_0^{\mathsf{dr}} = \tau$.

12.1 The doubly robust estimator

- By Thm. 12.1, **if either the propensity score model or the outcome model is correctly specified,** $\tilde{\mu}_1^{\mathsf{dr}} - \tilde{\mu}_0^{\mathsf{dr}} = \tau$.

Proof.

$$
\begin{aligned}
\tilde{\mu}_1^{\mathsf{dr}} - \mathbb{E}[Y(1)] &= \mathbb{E}\left[\frac{Z\{Y(1) - \mu_1(X, \beta_1)\}}{e(X, \alpha)} + \mu_1(X, \beta_1) - Y(1)\right] \\
&= \mathbb{E}\left[\frac{Z - e(X, \alpha)}{e(X, \alpha)}(Y(1) - \mu_1(X, \beta_1))\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\frac{Z - e(X, \alpha)}{e(X, \alpha)}(Y(1) - \mu_1(X, \beta_1)) \mid X\right]\right] \quad (\because \text{LIE}) \\
&= \mathbb{E}\left[\mathbb{E}\left[\frac{Z - e(X, \alpha)}{e(X, \alpha)} \mid X\right] \times \mathbb{E}\left[Y(1) - \mu_1(X, \beta_1) \mid X\right]\right] \quad (\because \text{strong ignorability}) \\
&= \mathbb{E}\left[\frac{e(X) - e(X, \alpha)}{e(X, \alpha)} \times \{\mu_1(X) - \mu_1(X, \alpha)\}\right].
\end{aligned}
$$

12.1 The doubly robust estimator

We can obtain their sample analogs to construct a doubly robust estimator for τ.

12.1 Definition (doubly robust estimator for the average causal effect)

Based on the data $(X_1, Z_i, Y_i)_{i=1}^n$, we can obtain a doubly robust estimator for τ by the following steps:

1. obtain the fitted values of the propensity scores: $e(X_i, \hat{\alpha})$;
2. obtain the fitted values of the outcome means: $\mu_1(X_i, \hat{\beta}_1), \mu_0(X_i, \hat{\beta}_0)$;
3. construct the doubly robust estimator: $\hat{\tau}^{dr} = \hat{\mu}_1^{\text{dr}} - \hat{\mu}_0^{\text{dr}}$, where

$$\hat{\mu}_1^{\text{dr}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{Z_i\{Y_i - \mu_1(X_i, \hat{\beta}_1)\}}{e(X_i, \hat{\alpha})} + \mu_1(X_i, \hat{\beta}_1)\right],$$

$$\hat{\mu}_0^{\text{dr}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{(1 - Z_i)\{Y_i - \mu_0(X_i, \hat{\beta}_0)\}}{1 - e(X_i, \hat{\alpha})} + \mu_0(X_i, \hat{\beta}_0)\right]$$

12.1 The doubly robust estimator

By definition, we can also write the doubly robust estimator as:

$$\hat{\tau}^{\text{dr}} = \hat{\tau}^{\text{reg}} + \frac{1}{n}\sum_{i=1}^{n} \frac{Z_i\{Y_i - \mu_1(X_i, \hat{\beta}_1)\}}{e(X, \hat{\alpha})} - \frac{1}{n}\sum_{i=1}^{n} \frac{(1 - Z_i)\{Y_i - \mu_0(X_i, \hat{\beta}_0)\}}{1 - e(X_i, \hat{\alpha})}.$$

We can also rewrite it as

$$\hat{\tau}^{\text{dr}} = \hat{\tau}^{\text{ipw}} - \frac{1}{n}\sum_{i=1}^{n} \frac{Z_i - e(X_i, \hat{\alpha})}{e(X_i, \hat{\alpha})}\mu_1(X_i, \hat{\beta}_1) + \frac{1}{n}\sum_{i=1}^{n} \frac{e(X_i, \hat{\alpha}) - Z_i}{1 - e(X_i, \hat{\alpha})}\mu_0(X_i, \hat{\beta}_0).$$

12.2 More intuition and theory for the doubly robust estimator

The IPW estimator for μ_1 based on $\mu_1 = \mathbb{E}\left[ZY/e(X)\right]$ completely ignores the outcome model of Y. With possibly misspecified outcome model $\mu_1(X, \beta_1)$,

$$\mu_1 = \mathbb{E}[Y(1)] = \mathbb{E}[Y(1) - \mu_1(X, \beta_1)] + \mathbb{E}[\mu_1(X, \beta_1)].$$

If we apply the IPW formula to the first term in the above formula:

$$\mu_1 = \mathbb{E}\left[\frac{Z(Y(1) - \mu_1(X, \beta_1))}{e(X)}\right] + \mathbb{E}[\mu_1(X, \beta_1)] = \mathbb{E}\left[\frac{Z(Y(1) - \mu_1(X, \beta_1))}{e(X)} + \mu_1(X, \beta_1)\right],$$

which holds if the propensity core model correct without assuming that the outcome model is correct.

Future Reading

- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. https://academic.oup.com/ectj/article/21/1/C1/5056401

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Scharfstein, D. O., A. Rotnitzky, and J. M. Robins (1999). Adjusting for nonignorable drop-out using semiparametric nonresponse models. *Journal of the American Statistical Association 94*(448), 1096–1120.